MRI Space Mission: A Stepwise Paediatric MRI Training Protocol for Young Children

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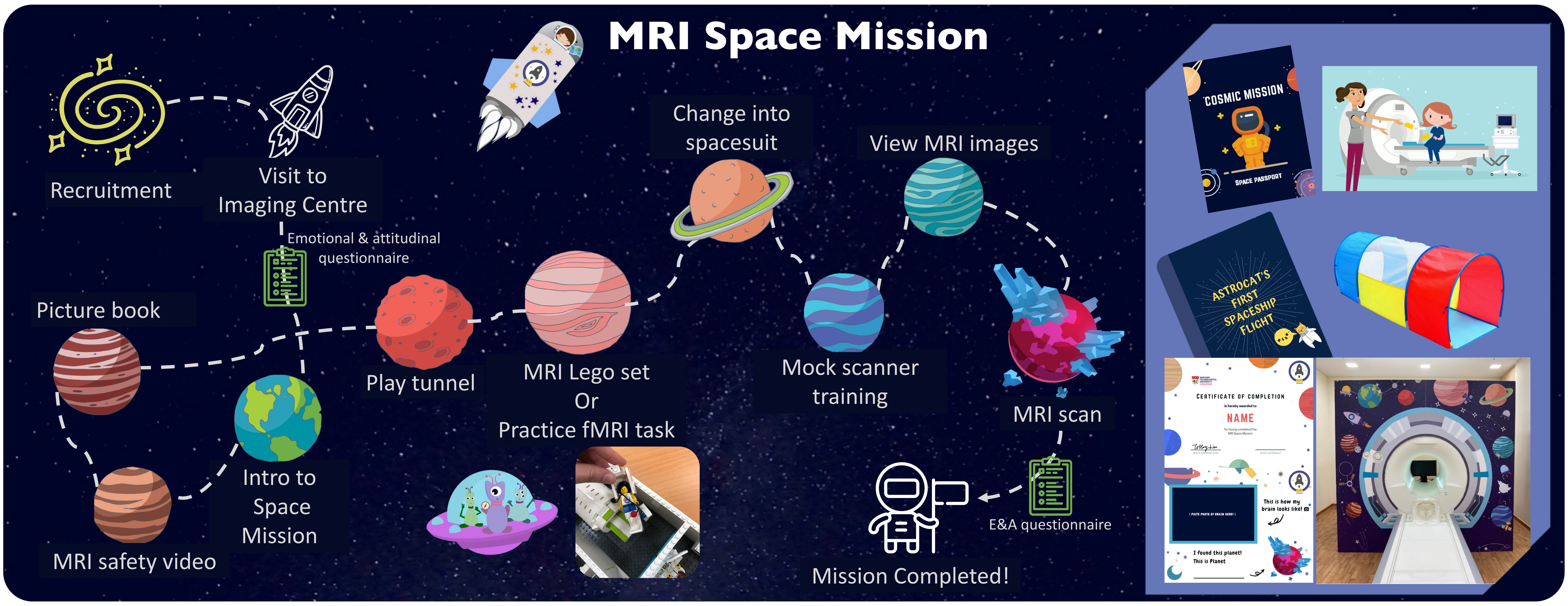
Introduction

- Magnetic Resonance Imaging (MRI) can be particularly challenging for young children, as it requires them to remain still for an extended period while lying inside a loud and enclosed scanner.
- Many children experience anxiety or fear when faced with the large, unfamiliar machine and the noises it produces [1].
- Movement during scanning often leads to motion artefacts, reducing image quality and sometimes necessitating scan repetition or sedation [2].
- Previous studies have shown that playful preparation can reduce anxiety and sedation rates in paediatric MRI [3,4].
- To address these challenges, we developed the **MRI Space Mission**, a child-centred, stepwise **paediatric MRI training protocol.**

Aim

- To improve scan success rates by familiarising children with the MRI environment in a playful way and reducing procedure-related anxiety.



Participants & MRI data

- 48 (5-7 y) and 40 (7-9 y) kindergarten and primary school children
- T1w (TR/TE = 2000/2.26); Functional (TR/TE=1500/30); Diffusion (TR/TE = 5000/91)
- 3T Siemens Prisma scanner, 32-channel head coil
- fMRI (SPM12 [6]): Motion detected by ARTrepair [7] (threshold = 0.6 mm; run excluded if >25% bad volumes). A dataset was considered successful if at least one language has >2/4 valid runs.
- DSI (DSI Studio [5]): Data were considered unsuccessful if neighbouring DWI correlation <0.7 or >15 bad slices.

Results

Sequence	Storyline video	Study 1 (5-7y, n=48)	Study 2 (7-9y, n=40)	Duration (mm:ss)
MRI scan completion rate		97.92%	100%	-
	Sound Check video	-	-	00:26
Localizer	Launching off video	-	-	00:14
T1	Cartoon	64.58%	82.5%	04:40
Field maps	Task/Water galaxy intro	-	-	01:24
Resting-state	Inscapes video	62.5%	N/A	07:10
fMRI	Language games (8 runs)	N/A	80%	21:20
T2	Arrival at mystery planet	70.83%	70%	02:14
DSI AP, PA	Continue cartoon	43.75%	45%	12:50
Total duration		27:52	42:42	Max 1 hr

Emotional & Attitudinal Ratings (5 points scale)		Study 1	Study 2
Emotional statements (child)	No significant pre–post changes		
Emotional statements (parent)*	after	secure↑; nervous↓ worried↓; unsure↓	worried↓ unsure↓
I know about the MRI spaceship (child)*	before	2(1.45)	2.13(1.4)
	after	4.21(1.1)	4.51(0.72)
I'm familiar with what the MRI machine does (parent)*	before	3.54(1.04)	3.17(0.89)
	after	4.52(0.69)	4.12(0.6)
I feel uncertain about my child going through an MRI scan (parent)*	before	2.38(1.11)	2.23(0.88)
	after	0.95(1.81)	1.6(0.61)

Paired t-tests were done for all items (before vs. after). *p < 0.0036, Bonferroni corrected.

Conclusions

- The 98.86% attendance rate and the high activity ratings indicated that our MRI training protocol effectively increased both children's and parents' willingness to participate in an MRI study.
- Overall, our completion rates were higher than those reported in previous studies, and our sequence-level success rates were comparable to or better than prior paediatric MRI work that implemented child-friendly preparation [2,3,8,9].
- Our space-themed MRI training protocol is feasible and effective.
- However, the results suggest that there is still room to further reduce head motion and improve children's ability to remain still with additional measures.

Scan to watch

References

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